Zed Block Examples

Example 1 : Basic Variable Declarations

Use axdef for variable declarations with types:

$$[Person, Department]$$

$$\begin{array}{|l}
assignment : Person \pfun Department \\
employees : \mathbb{F}\ Person \\
departments : \mathbb{F}\ Department
\end{array}$$

This groups related variable declarations in a single block (given types must be declared outside).

Example 2 : Abbreviations

$$N\_PLUS == \{\, n : \mathbb{N} \mid n > 0 \,\}$$

$$N\_EVEN == \{\, n : \mathbb{N} \mid n \bmod 2 = 0 \,\}$$

$$N\_ODD == \{\, n : \mathbb{N} \mid n \bmod 2 = 1 \,\}$$

Abbreviations can be defined standalone (zed blocks for abbreviations may not be fully supported yet).

Example 3 : Type Parameters

$$[Company]$$

Declare given types using the given keyword.

Example 4 : Predicates elem Zed Block

$$\begin{array}{|l}
x : \mathbb{N} \\
y : \mathbb{N} \\
\hline
x > 0 \\
y > 0 \\
x + y < 100
\end{array}$$

Multiple predicates can constrain variables defined in axdef.

Example 5 : Free Type Definitions

$$Color ::= red \mid green \mid blue$$

$$Size ::= small \mid medium \mid large$$

$$\begin{array}{|l}
\multicolumn{1}{l}{\underline{\ Product\ }} \\
color : Color \\
size : Size \\
price : \mathbb{N} \\
\hline
price > 0 \\
\hline
\end{array}$$

Free types are defined standalone, then used in schemas.

Example 6 : Multiple Schemas

$[AccountZB, BalanceZB]$

BankAccountZB

$$accountNumber : \mathbb{N}$$
$$balance : \mathbb{Z}$$

$$balance \geq 0$$

TransactionZB

$$from : BankAccountZB$$
$$to : BankAccountZB$$
$$amount : \mathbb{N}$$

$$amount > 0$$
$$from.balance \geq amount$$

Multiple schemas can be defined separately, each with their own constraints.

Example 7 : Schema with External Variables

$[Student, Course]$

$$enrolled : Student \leftrightarrow Course$$
$$maxCourses : \mathbb{N}$$
$$maxStudents : \mathbb{N}$$

$$maxCourses = 7$$
$$maxStudents = 500$$

Enrollment

$$students : \mathbb{F}\ Student$$
$$courses : \mathbb{F}\ Course$$

$$students = \mathrm{dom}\ enrolled$$
$$courses = \mathrm{ran}\ enrolled$$
$$\#students \leq maxStudents$$
$$\#courses \leq maxCourses$$

Schemas can reference variables defined in axdef blocks.

Example 8 : Nested Schemas

$[Char]$

Address

$$street : \mathrm{seq}\ Char$$
$$city : \mathrm{seq}\ Char$$
$$zipCode : \mathbb{N}$$

$$zipCode \geq 10000 \land zipCode \leq 99999$$

$$
\begin{array}{|l}
\hline
\multicolumn{1}{l}{\textit{PersonData}} \\
name : \mathrm{seq}\ Char \\
address : Address \\
age : \mathbb{N} \\
\hline
age \geq 0 \land age \leq 150 \\
\hline
\end{array}
$$

$$
\begin{array}{|l}
population : \mathbb{F}\ PersonData
\end{array}
$$

Schema Address is used within schema PersonData. Variables using schemas go in axdef.

Example 9 : Constants land Functions

$$PI == 3$$

$$E == 2$$

$$
\begin{array}{|l}
circumference : \mathbb{N} \longrightarrow \mathbb{N} \\
area : \mathbb{N} \longrightarrow \mathbb{N} \\
\hline
\forall r : \mathbb{N} \bullet circumference(r) = 2 * PI * r \\
\forall r : \mathbb{N} \bullet area(r) = PI * r * r
\end{array}
$$

Mathematical constants defined as abbreviations, functions in axdef.

Example 10 : Permission System Example

$$[UserID, DocumentID]$$

$$PermissionType ::= read \mid write \mid admin$$

$$
\begin{array}{|l}
\hline
\multicolumn{1}{l}{\textit{User}} \\
userId : UserID \\
documents : \mathbb{P}\ DocumentID \\
permissions : DocumentID \rightarrowtail\!\!\!\!\!+ PermissionType \\
\hline
\mathrm{dom}\ permissions \subseteq documents \\
\hline
\end{array}
$$

$$
\begin{array}{|l}
\hline
\multicolumn{1}{l}{\textit{System}} \\
users : UserID \rightarrowtail\!\!\!\!\!+ User \\
allDocuments : \mathbb{F}\ DocumentID \\
\hline
allDocuments = \bigcup \{ u : \mathrm{ran}\ users \bullet u.documents \} \\
\hline
\end{array}
$$

$$
\begin{array}{|l}
users : UserID \rightarrowtail\!\!\!\!\!+ User
\end{array}
$$

A permission system specification using free types and schemas.

Example 11 : Best Practices

Use zed blocks for: 1. Type parameter declarations [X, Y] 2. Abbreviations (N_PLUS, N_EVEN, etc.) 3. Simple predicates without variable declarations

Use axdef for: 1. Variable declarations with types 2. Constraints on those variables 3. Function definitions

Use schema for: 1. State spaces with declarations 2. Operations on state 3. Reusable component specifications